Multimodal RAG Pipeline

Retrieval augmented generation combines document retrieval with text generation.
The ingestion service parses PDF files into text blocks, tables and figures.
Kafka decouples upload from parsing so the HTTP request returns immediately.
MinerU performs layout analysis and extracts figures with captions.
The retrieval layer uses BM25 by default and can be upgraded to dense vectors.
Answers are rendered as Markdown, so figures can appear inline with the text.